



Optimal planning of renewable-based mining microgrids: a comparative study of multi-objective evolutionary algorithms

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Abstract

The design of renewable-powered mining microgrids is an essential study for the shift towards electric and carbon-neutral operations within the mining sector. This paper presents a multi-objective two-stage optimisation framework for the planning of microgrids in remote mine sites. The first-stage (strategic) problem aims to minimise the total net present cost, total greenhouse gas emissions, renewable energy curtailment, and improve system reliability. In the second-stage (scheduling) problem, the energy scheduling of the microgrid's energy sources is determined using a rule-based approach. The proposed framework provides optimal sizing for renewable energy sources, energy storage systems, and fossil-fuel backup generators to meet the microgrid's electricity demand, demonstrating the feasibility and benefits of renewable-based microgrid deployment in mining operations. To solve this complex problem, several state-of-the-art multi-objective evolutionary algorithms, including the non-dominated sorting genetic algorithm II (NSGA-II), the NSGA-III, the unified NSGA-III (U-NSGA-III), S-metric selection evolutionary multi-objective algorithm (SMS-EMOA), and the adaptive geometry estimation multi-objective evolutionary algorithm (AGE-MOEa) and its newer version AGE-MOEa-II, are applied to efficiently explore the performance of these techniques in finding the best trade-off between competing objectives. The effectiveness of these algorithms is evaluated and compared in the context of a real-world mining case study. The simulation results demonstrate that SMS-EMOA and AGE-MOEa outperform other algorithms in terms of convergence and diversity, particularly for complex microgrid configurations. The study highlights the potential of renewable-based microgrids to significantly reduce emissions, with trade-offs in cost, depending on the inclusion of energy storage solutions.

Keywords Hybrid energy systems · Renewable energy sources · Multi-objective optimisation · Two-stage programming · Mining industry · Microgrid · Energy storage system

Extended author information available on the last page of the article

Nomenclature

i, Ψ	Index and set of microgrid energy sources
t, T	Index and set of time intervals
k, K^i	Index and set of components of energy source i
y	Index of year
ΔT	Time interval duration
Pd_t	Amount of load demand at time t
L_{mg}	Microgrid lifetime
em_{DG}	Per unit emission of diesel generator
u_{L-GJ}	Unit conversion factor from litres to gigajoules
c_k	Per unit capital cost of component k
S_{in}, η_{in}, pf	Inverter rated apparent power, efficiency, and power factor
P_{md}	Rated power of each PV module
$R_{DC/AC}$	DC/AC ratio of solar PV system
l_{md}, w_{md}	Length and width of PV module
α_O, γ_{is}	Solar PV system operational and maintenance (O&M) and insurance cost factors, respectively
f_{PV}, W_{md}	Failure rate of PV modules and their warranty period in years
ir	Compound interest rate
L_k	Lifetime of component k
$c_{u,O}^{WT/BT}$	Operational and maintenance (O&M) cost of wind/battery systems
EoL	End-of-life fraction of battery
c_{DG}^I	Per unit capital cost of diesel generator
c_{DG}^M, c_{DG}^F	Maintenance and fuel cost factors of diesel generator
a_{DG}, b_{DG}	Fuel consumption factors of the generators in litres per energy unit
ENS	Maximum energy-not-served allowed
$c_{rate}^{min}, c_{rate}^{max}$	Minimum and maximum c-rating of battery
P_i^{max}	Maximum installed power capacity of energy source i
$\eta_t^{ch}, \eta_t^{dis}$	Charging and discharging efficiency of the battery
P_t^{pv}, P_t^{wt}	Predicted normalised solar and wind power generation at time t
$\alpha^{min/max}$	Minimum/maximum limits of state of charge of the battery
l_{yr}^{DG}	Total diesel fuel consumption in litres per year
P_t^{ns}	Power not served at time t
P_t^{pcurt}	Renewable power curtailment at time t
P_{WT}, P_{DG}	Installed power capacity of wind system and diesel generator
P_{DC}^{PV}, P_{AC}^{PV}	DC and AC installed power capacity of solar PV system
P_{BT}, E_{BT}	Installed power and energy capacity of battery
$e_{loss, yr}$	The percentage of energy loss of the battery
P_t^{dis}, P_t^{ch}	Discharging and charging power of the battery at time t
e_t	State of energy of the battery at time t

1 Introduction

The mining industry is an energy-intensive sector highly dependent on fossil fuels, both in its early development stages and throughout its operational life. Energy consumption within the mining industry accounts for approximately 38% of all industrial energy use and about 11% of global energy consumption [1]. Additionally, the mining industry contributes 4%-7% to global greenhouse gas (GHG) emissions [2]. This substantial energy consumption is projected to double by 2050 without the implementation of new policy measures [3]. At the mine level, energy costs comprise up to 30% of the operational expenses of a mine [4], which is substantial and leads to higher prices for commodities. Consequently, the mining industry is actively moving towards decarbonisation and electrification through the integration of renewable energy sources (RES) and battery electric vehicles (BEVs) to reduce fossil fuel consumption and GHG emissions, driven by environmental regulations and social pressures [1, 5, 6]. However, incorporating large-scale RES in the mining microgrid (MG), which frequently operates independently of the main grid, introduces complexities in managing variable load profiles and ensuring compatibility with the intermittent nature of RES generation [1, 5]. Hence, an optimised MG consisting of a high share of RES, battery energy storage systems (BESSs) and fossil-fuel backup generators is crucial to provide uninterrupted power supply and meet the energy demand while minimising the energy cost.

Numerous studies have developed various MG design frameworks for typical residential, commercial, and community MGs, both in grid-connected and off-grid modes, to minimise total costs and emissions. For instance, [7] and [8] present frameworks for designing off-grid residential MGs consisting of solar, wind, BESSs, and diesel generators to meet the demand of rural areas in Algeria and Morocco, in order to reduce the total cost of electricity. In [9] and [10], multi-objective optimisation models are proposed for sizing grid-connected MGs to minimise both total costs and emissions.

However, there is a limited body of literature on MG applications within the mining industry. In [11], a framework for optimal sizing of off-grid microgrids is introduced to evaluate the profitability of renewable energy systems, with its effectiveness demonstrated across three mining case studies in Australia. The authors in [12] performed a techno-economic analysis to evaluate the feasibility of integrating concentrated solar power with thermal storage and solar photovoltaic systems combined with battery storage to provide energy to a typical mine in Zimbabwe. Ref. [13] presents a multi-objective model for optimal sizing of grid-connected hybrid MGs consisting of solar-biogas RES and energy storage systems to minimise total costs and GHG emissions in mine sites. The work in [3] evaluates the economic potential of hybrid MGs in mining using the HELIOS-Mining model, considering technical, economic, and spatial factors. It focused on grid-connected and off-grid mines in Northern Chile and off-grid mines in North-Western Australia and the Yukon, Canada.

Despite the valuable insights these studies offer on the integration of RES in MG design and planning, few works focus on heavy industries such as mining in

remote areas, where the MG must handle a substantial, high-quality, and typically continuous energy supply, often operating 24/7, posing significant challenges for the large-scale integration of RES [14]. For example, wind and solar farms may generate surplus energy during certain hours, leading to energy curtailment and potential financial losses. Although BESSs can store a portion of this energy, current battery technology frequently provides limited storage capacity at affordable costs. Additionally, obtaining net metering or export compensation can be challenging for mining projects in remote areas due to regulatory restrictions, transmission system power limitations, and low compensation rates [1, 14].

To address these research gaps, this paper introduces a MG planning model for the mining industry with high penetration of RES. The model is formulated as a multi-objective two-stage optimisation problem to determine the optimal capacity of solar and wind systems, BESS units, and fossil-fuel backup generators to meet MG demand. In the first-stage (strategic) problem, the objective is to minimise the total net present cost (NPC) of the MG, the total GHG emissions, the total curtailment of RES, and to improve the reliability of the system, while accounting for both the calendar and cycling degradation of BESSs. The second-stage (scheduling) problem models the operation of the MG using a rule-based approach to schedule the output of each component in the MG to meet the demand efficiently. To solve this problem, several advanced multi-objective evolutionary algorithms (MOEAs) are employed, including the non-dominated sorting genetic algorithm II (NSGA-II), the NSGA-III, the unified NSGA-III (U-NSGA-III), the S-metric selection evolutionary multi-objective algorithm (SMS-EMOA), and the adaptive geometry estimation multi-objective evolutionary algorithm (AGE-MOEa) and its newer version AGE-MOEa-II. Their effectiveness in exploring the trade-offs between conflicting objectives is evaluated and discussed. The key contributions of this paper are summarised as follows:

- A novel multi-objective two-stage optimisation model is proposed for optimally sizing MG components at remote mining locations, incorporating wind, solar, BESSs, and fossil-fuel backup generators, with consideration of a comprehensive battery degradation model.
- The proposed model is solved using several state-of-the-art multi-objective evolutionary algorithms to identify Pareto-optimal solutions and analyse their effectiveness in capturing the trade-offs between economic, environmental, and reliability objectives.
- A real-world MG case study is used for a remote underground mine to validate the effectiveness of the proposed model, using historical load data, anticipated renewable energy generation, and details of commodity and capital prices.
- The implemented framework is available on <https://bit.ly/48xfm6X>. The framework includes the multi-objective evolutionary algorithms used in this study and also includes instructions on how to run the code. It offers transparency and accessibility, enabling researchers and practitioners to seamlessly replicate, adapt, and expand upon the current work.

The remainder of this paper is organised as follows: Sect. 2 provides an overview of the proposed MG planning model. Section 3 delves into the detailed problem formulation for both the strategic and scheduling problems. Section 4 presents the numerical results and a comprehensive discussion. Finally, Sect. 5 concludes the paper with insights and recommendations for future research directions.

2 Model overview

Figure 1 presents a high-level diagram of the proposed multi-objective, two-stage framework for MG planning in mining operations. As shown in the figure, the design framework incorporates various site-specific input data, including RES production at the mine site, electricity load demand, and financial parameters. It also requires technology-specific data related to each DER technology, particularly cost and emission data and the battery degradation model. For this analysis, the Battery Lifetime Analysis and Simulation Toolsuite (BLAST) developed by NREL is used for the end-of-life of the BESS evaluation [15].

The design framework incorporates a two-stage optimisation problem where the first-stage problem involves multi-objective optimisation, aiming to minimise the NPC, GHG emissions, RES curtailment, and energy-not-served (ENS) as a reliability index. This stage determines the optimal combination and sizing of DERs, including PV, wind, BESS, and backup generators. The second-stage problem then solves the economic dispatch problem of the MG based on the mix and sizing of the MG component in each feasible solution.

The output of this design framework consists of Pareto optimal solutions and trade-offs based on the specified objectives. These solutions represent non-dominated solutions for the MG design that offer various sizing scenarios for DER technologies, providing decision-makers with a range of viable options to consider.

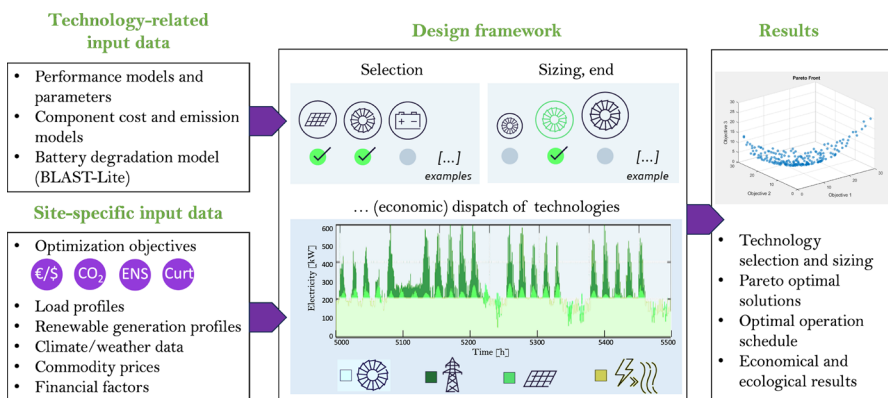


Fig. 1 The structure of the proposed two-stage MG design model

3 Mathematical formulation

This section provides a detailed formulation of the proposed MG design model. First, the mathematical formulation of the strategic problem is outlined, including the cost and emission models associated with the MG components. Subsequently, the MG scheduling problem, also referred to as the scheduling model, is introduced.

3.1 Strategic problem

The minimisation objective functions of the strategic problem, are expressed as:

$$LCOE = \frac{\sum_{i \in \Psi} (C_i^C + C_i^O + C_i^R - C_i^S)}{\sum_{t \in T} Pd_t \cdot \Delta T \cdot L_{mg}} \quad (1)$$

$$EM_{pu} = \frac{l_{yr}^{DG} \cdot u_{L-GJ} \cdot em_{DG}}{\sum_{t \in T} Pd_t \cdot \Delta T} \quad (2)$$

$$Curt = \sum_{t \in T} P_t^{Curt} \cdot \Delta T \quad (3)$$

$$ENS = \sum_{t \in T} P_t^{ns} \cdot \Delta T \quad (4)$$

where (1) minimises the levelised cost of energy (LCOE) of the MG, which is calculated by dividing the total NPC over the life of the MG by total energy demand. In this equation, T is the set of time intervals in a year, indexed by t and length of ΔT , while $\Psi = \{PV, WT, BT, DG\}$ is the set of DERs in the MG consisting of solar PV, wind turbines, lithium-ion BESS, and diesel generators. Also, $C_i^{[.]}$ denotes the present value of the cost associated with each energy source i and the superscripts C , O , R , and S represent capital, operation and maintenance, replacement, and salvage, respectively. Parameters L_{mg} and Pd_t represent the MG lifetime in years and the power demand at each time interval t .

Furthermore, (2)–(4) compute the normalised value of operational GHG emissions (EM_{pu}), total RES curtailment ($Curt$), and energy-not-served (ENS) as the reliability index, respectively. In these equations, l_{yr}^{DG} represents the total diesel fuel consumption of the generators in litres per year, while P_t^{Curt} and P_t^{ns} are the RES energy curtailment and the energy-not-served in each time interval t , respectively, which are calculated by solving the scheduling problem. In addition, parameters em_{DG} and u_{L-GJ} are the normalised value of the emissions of the diesel generator g and the unit conversion factor from litres to gigajoules.

The cost and emission models of the DERs and their mathematical representation are provided in the following subsections.

3.1.1 Solar PV system

The total capital cost of a PV system, expressed as C_{PV}^C , consists of the cost of modules (md), inverters (in), land (ld), structures (st), cables (cb), tracking system (ts) and transformers (tr) as:

$$C_{PV}^C = \sum_{k \in K^{PV}} C_k \quad (5)$$

where $K^{PV} = \{\text{md}, \text{in}, \text{ld}, \text{st}, \text{cb}, \text{ts}, \text{tr}\}$. The capital costs of modules and inverters are associated with the number of PV modules and inverters that can be calculated as:

$$C_k = c_k \cdot N_k; \forall k \in \{\text{md}, \text{in}\} \quad (6)$$

where c_k is the per unit cost of modules (md) and inverters (in), respectively. N_k is the number of modules and inverters that are calculated as:

$$N_{\text{md}} = \left\lceil \frac{P_{DC}^{PV}}{P_{\text{md}}} \right\rceil; N_{\text{in}} = \left\lceil \frac{P_{AC}^{PV}}{S_{\text{in}} \cdot pf} \right\rceil; P_{AC}^{PV} = \eta_{\text{in}} \cdot \left(\frac{P_{DC}^{PV}}{R_{\frac{DC}{AC}}} \right) \quad (7)$$

where P_{DC}^{PV} and P_{AC}^{PV} are the nominal DC and AC power capacities of the installed solar PV system, respectively; P_{md} is the power capacity of each PV module; η_{in} is the efficiency of inverters, $R_{\frac{DC}{AC}}$ is the DC/AC ratio of solar PV system, S_{in} is the rated apparent power of each inverter, and pf is the inverter rated power factor. The total cost of land required to install PV modules (C_{ld}) is calculated as:

$$C_{\text{ld}} = (c_{\text{ld}}^u + c_{\text{pt}}^u) \times (\alpha_{\text{ld}} \cdot (N_{\text{md}} \cdot l_{\text{md}} \cdot w_{\text{md}})) \quad (8)$$

where c_{ld}^u and c_{pt}^u are the cost of land lease and lightning protection, respectively. l_{md} and w_{md} are module dimensions, and α_{ld} is the area extension factor allowing additional space needed for inverters, transformers, cable channels, roads, ditches, and the control building. Other cost terms, i.e. the cost of the structure and tracking system, AC and DC cables, and transformers are calculated as follows:

$$C_k = c_k \cdot P_{DC}^{PV}; \forall k \in \{\text{st}, \text{ts}\} \quad (9)$$

$$C_{\text{cb}} = \sum_{k \in \{\text{DC}, \text{AC}\}} l_k^u \cdot P_k^{PV} \cdot c_k; \quad (10)$$

$$C_{\text{tr}} = c_{\text{tr}} \cdot P_{AC}^{PV} \quad (11)$$

where $c_k, \forall k \in \{st, ts, DC, AC, tr\}$ is the per unit cost of structures, tracking systems, DC and AC cables, and transformers while l_{DC}^u and l_{AC}^u are the per unit DC and cable length cost, respectively.

Generally, the operation and maintenance cost of a PV system is calculated as a portion of its annualised capital cost [16]. However, in our cost model, another term related to annual insurance payment is added to make it more aligned with practice. These costs are then converted to the present value using the compound interest rate (ir) and the MG's life (L_{mg}) to be part of the NPC. Thus, the present value of the operation cost, C_{PV}^O , can be expressed as follows:

$$C_{PV}^O = \sum_{y=1}^{L_{mg}} \frac{\alpha_O \cdot C_{PV}^C}{(1+ir)^y} + \sum_{y=1}^{L_{mg}} \frac{\gamma_{is} \cdot C_{PV}^C}{(1+ir)^y} \quad (12)$$

where α_O and γ_{is} are the operational and maintenance (O&M) and insurance cost factors, respectively, and y is the year count.

Furthermore, during the MG life, we may need to replace several assets, such as inverters and PV modules. In addition, as the MG reaches the end of its lifespan, various assets contribute salvage values that must be accounted for in the NPC. Hence, the present value of the replacement cost (C_{PV}^R) and salvage value (C_{PV}^S) of the PV system can be expressed as:

$$C_{PV}^R = \sum_{y=1}^{\lfloor \frac{L_{mg}}{L_{in}} \rfloor} \frac{C_{in}}{(1+ir)^{y \cdot L_{in}}} + \sum_{y=Wr_{md}+1}^{L_{mg}} (f_{PV} \cdot N_{md}) \cdot \frac{C_{md}}{(1+ir)^y} \quad (13)$$

$$C_{PV}^S = \sum_{k \in K^{PV}} C_k \cdot \left(1 - \frac{2}{L_k}\right)^{L_{mg}} \cdot d; \forall k \in \{md, in, ld, st, cb, ts, tr\} \quad (14)$$

where L_{in} and Wr_{md} are, respectively, the inverter life and the module warranty period. L_k represents the set of lifetimes of the PV system components, while f_{PV} is the failure rate of PV modules and d is the discount factor that is calculated as $d = \frac{1}{(1+ir)^{L_{mg}}}$ based on the MG's life. It is worth mentioning that these replacement costs assume that the life of the PV system (L_{PV}) is more than the life of the mine (L_{mg}). However, the developed replacement cost model can easily be modified for cases that $L_{mg} > L_{PV}$.

3.1.2 Wind system

Similarly, the capital cost of the wind system includes three terms, that are, turbine cost (C_{WT}^{tb}), balance of system (C_{WT}^{bs}), and construction funding and contingencies (C_{WT}^{cn}), which are calculated as follows:

$$C_{WT}^C = C_{WT}^{tb} + C_{WT}^{bs} + C_{WT}^{cn} \quad (15)$$

$$C_{WT}^{tb} = \sum_{k \in K^{tb}} c_k \cdot P_{WT}; \forall k \in \{rt, nc, tw\} \quad (16)$$

$$C_{WT}^{bs} = \sum_{k \in K^{bs}} c_k \cdot P_{WT}; \forall k \in \{eg, pm, fd, sf, it, ei\} \quad (17)$$

$$C_{WT}^{cn} = \sum_{k \in K^{cn}} c_k \cdot P_{WT}; \forall k \in \{cf, cy\} \quad (18)$$

where P_{WT} is the nominal power capacity of the wind system and c_k is the per unit costs for $K^{WT} = (K^{tb} \cup K^{bs} \cup K^{cn})$, which includes the rotor (rt), nacelle (nc) and tower (tw) of turbines; engineering (eg), project management (pm), foundation (fd), site access and facilities (sf), installation (it) and electrical infrastructure (ei) of balance of system; and finally construction funding (cf) and contingency (cy).

The present values of the operational and replacement costs, as well as the salvage value of the wind system, are calculated as follows considering that only the rotor and the nacelle need to be replaced during the life of the MG and the turbine consisting of rotor, nacelle, and tower is the only valuable asset with salvage value.

$$C_{WT}^O = \sum_{y=1}^{L_{mg}} \frac{P_{WT} \cdot c_{u,O}^{WT}}{(1+ir)^y} \quad (19)$$

$$C_{WT}^R = \sum_{y=1}^{\lfloor \frac{L_{mg}}{L_{rt}} \rfloor} \frac{C_{rt}}{(1+ir)^{y \cdot L_{rt}}} + \sum_{y=1}^{\lfloor \frac{L_{mg}}{L_{nc}} \rfloor} \frac{C_{nc}}{(1+ir)^{y \cdot L_{nc}}} \quad (20)$$

$$C_{WT}^S = C_{WT}^{tb} \cdot \left(1 - \frac{2}{L_{WT}}\right)^{L_{mg}} \cdot d \quad (21)$$

where $c_{u,O}^{WT}$ represents the normalised cost of operation and maintenance per year, while L_{rt} , L_{nc} , and L_{WT} are the lifetimes of the rotor, nacelle and wind turbine, respectively.

3.1.3 Battery energy storage system (BESS)

The capital cost of BESS includes two main components associated with energy capacity and power as:

$$C_{BT}^C = C_{BT}^e + C_{BT}^p \quad (22)$$

$$C_{BT}^e = \sum_{k \in K^e} c_k \cdot E_{BT}; \forall k \in \{ec, bs, si, ep, pd\} \quad (23)$$

$$C_{BT}^p = \sum_{k \in K^p} c_k \cdot P_{BT}; \forall k \in \{pc, cc\} \quad (24)$$

where E_{BT} and P_{BT} are the energy and power capacity of the BESS, and c_k represents the per unit costs of $K^{BT} = (K^e \cup K^p)$, which includes the energy-related components including energy capacity (ec), balance of system (bs), system integration (si), engineering and procurement (ep), project development (pd), as well as the power-related per unit cost, which includes the power conversion system (pc) and control and communication (cc).

Similar to RES units, the present value of the operation and maintenance cost of BESS can be expressed as follows:

$$C_{BT}^O = \sum_{y=1}^{L_{mg}} \frac{P_{BT} \cdot c_{u,O}^{BT}}{(1+ir)^y} \quad (25)$$

where $c_{u,O}^{BT}$ is the normalised cost of operation and maintenance per year in terms of $\frac{\$}{kW-yr}$.

Generally, the replacement cost for BESSs consists of two parts: (1) power conversion system and (2) energy unit. The replacement cost of the power conversion system is similar to the PV system. However, the replacement cost of the energy units is related to the degradation of battery capacity due to cycling and calendar ageing. To this end, the present value of the replacement cost of BESS (C_{BT}^R) is as follows.

$$C_{BT}^R = \sum_{y=1}^{\lfloor \frac{L_{mg}}{L_{PCS}^{BT}} \rfloor} \frac{C_{pc}^{BT}}{(1+ir)^{y \cdot L_{PCS}^{BT}}} + \sum_{y=1}^{\lfloor \frac{L_{mg}}{L_e^{BT}} \rfloor} \frac{C_{ec}^{BT}}{(1+ir)^{y \cdot L_e^{BT}}} \quad (26)$$

where C_{pc}^{BT} and C_{ec}^{BT} represent the cost of the power conversion system and the energy storage system, respectively, and L_{PCS}^{BT} is the life of the power conversion system. Additionally, L_e^{BT} represents the life of the battery energy units that can be calculated based on calendar and cycling ageing. In this work, we use the BLAST lite library to calculate battery degradation [15]. According to this package, L_e^{BT} can be expressed as:

$$L_e^{BT} = \frac{1 - EoL}{e_{loss, yr}} \quad (27)$$

$$e_{loss, yr} = f_{BLAST}(SOC, Temp, BT_{Type}) \quad (28)$$

where EoL is the end-of-life fraction of the battery and $e_{loss, yr}$ represents the fraction of energy loss of the battery per year that is calculated based on the state of charge

(SOC) profile of battery, temperature, and battery type using the BLAST-lite package [15].

The salvage value of BESS (C_{BT}^S) is related to both energy and power components as:

$$C_{BT}^S = \sum_{k \in \{ec, bs, pc\}} C_k^{BT} \cdot \left(1 - \frac{2}{L_k^{BT}}\right)^{L_{u,k}^{BT}} \cdot d \quad (29)$$

where L_k^{BT} and $L_{u,k}^{BT} \forall k \in \{ec, bs, pc\}$ denote the useful life of the component k and the elapsed time in years since the installation of the most recent component, respectively.

3.1.4 Backup generators

Generally, the backup energy source in an off-grid MG is a diesel generator. Similarly to other technologies, the NPC of diesel generators includes the capital and operational costs as well as the salvage value, as:

$$C_{DG}^C = P_{DG} \cdot c_{DG}^I \quad (30)$$

$$C_{DG}^O = \sum_{y=1}^{L_{mg}} \frac{P_{DG} \cdot c_{DG}^M + l_{yr}^{DG} \cdot c_{DG}^F}{(1 + ir)^y} \quad (31)$$

$$C_{DG}^S = C_{DG}^C \cdot \left(1 - \frac{2}{L_{DG}}\right)^{L_{mg}} \cdot d \quad (32)$$

where P_{DG} and L_{DG} represent the installed power capacity and the useful life of the diesel generators, while c_{DG}^I , c_{DG}^M , and c_{DG}^F are the normalised capital, non-fuel operation & maintenance cost of the diesel generator, as well as the normalised cost of diesel fuel, respectively. Moreover, l_{yr}^{DG} represents the annual diesel fuel consumption that is calculated based on power generation as:

$$l_{yr}^{DG} = \sum_{t \in T} (a_{DG} \cdot P_t^{DG} + b_{DG} \cdot u_t^{DG}) \Delta T \quad (33)$$

where P_t^{DG} is the power required from the diesel generators at time t with the time interval ΔT , while u_t^{DG} represents the on-off status of the diesel generators at time t . Also, a_{DG} , b_{DG} are the fuel consumption factors of the generators in litres per energy unit. The values of P_t^{DG} are the solutions obtained from the scheduling problem.

The strategic problem is subject to the following constraints with respect to the maximum installed capacity of DERs (34) and the relationship between BESS power and energy capacities (35) as:

$$0 \leq P_i \leq P_i^{max} \quad \forall i \in \Psi \quad (34)$$

$$C_{rate}^{min} \cdot E_{BT} \leq P_{BT} \leq C_{rate}^{max} \cdot E_{BT} \quad (35)$$

where c_{rate}^{min} and c_{rate}^{max} are the minimum and maximum installed c-rate of BESS, respectively, while P_i^{max} represents the maximum installed power capacity of DER $i \in \Psi$. Moreover, ENS in the design problem is usually forced to be less than a certain value [17]. Hence, in this work, (4) is converted to a constraint as follows:

$$\sum_{t \in T} P_t^{ns} \cdot \Delta T \leq \overline{ENS} \quad (36)$$

where $\overline{ENS}$ is the maximum allowable of energy not served.

3.2 Scheduling model

The second-stage problem (scheduling model) simulates the operation of the MG and determines the power dispatch of each DER to meet the demand for a given design plan based on a predefined set of operational rules. This approach ensures that the MG meets the demand efficiently while minimising reliance on fossil-fuel generators. It should be noted that in this work, we have not explicitly considered the uncertainty of RES, hence uncertainty has not been considered in the optimisation problem. The main reason for that is to focus primarily on assessing the potential of evolutionary algorithms for the planning of mining MGs. However, the scheduling algorithm is executed continuously for a full year using real RES data. As a result, different possible combinations of RES, BESS, and load variations are considered in the optimisation problem, enabling the model to adapt to potential fluctuations. This approach essentially mimics robust optimisation under uncertainty.

The scheduling follows a hierarchical priority order:

1. **RES Utilisation:** The primary objective is to use all available RES generation (solar and wind) to meet the demand.
2. **Energy Storage Management:** Any excess RES generation is stored in the BESS, provided that there is available capacity. If the BESS reaches its maximum state of charge, the surplus energy is curtailed.
3. **Battery Discharge:** If the RES output is insufficient to meet the demand, the stored energy in the BESS is used to bridge the gap.
4. **Backup Generation:** If neither RES nor BESS can fully satisfy the demand, fossil-fuel generators are activated as a last resort to ensure a continuous power supply.
5. **Energy Not Served:** If, after exhausting all integrated energy resources, the available cumulative power does not satisfy the demand, the deficit will result in an energy shortfall, which causes an ENS event.

This rule-based approach is particularly relevant to the design of mining MGs, which aim to minimise reliance on fossil fuels and reduce GHG emissions while ensuring a stable and reliable power supply. In this regard, the scheduling model prioritises renewable generation to reduce operational costs and emissions while

maintaining the reliability of the MG. Incorporating this scheduling strategy into the case study takes into account practical operational restrictions, including limitations on the charge and discharge of the BESS, constraints on generator start-up, and projected availability of RES.

Fig. 2 details the flowchart of the proposed rule-based scheduling approach, offering an enhanced understanding of the decision-making process.

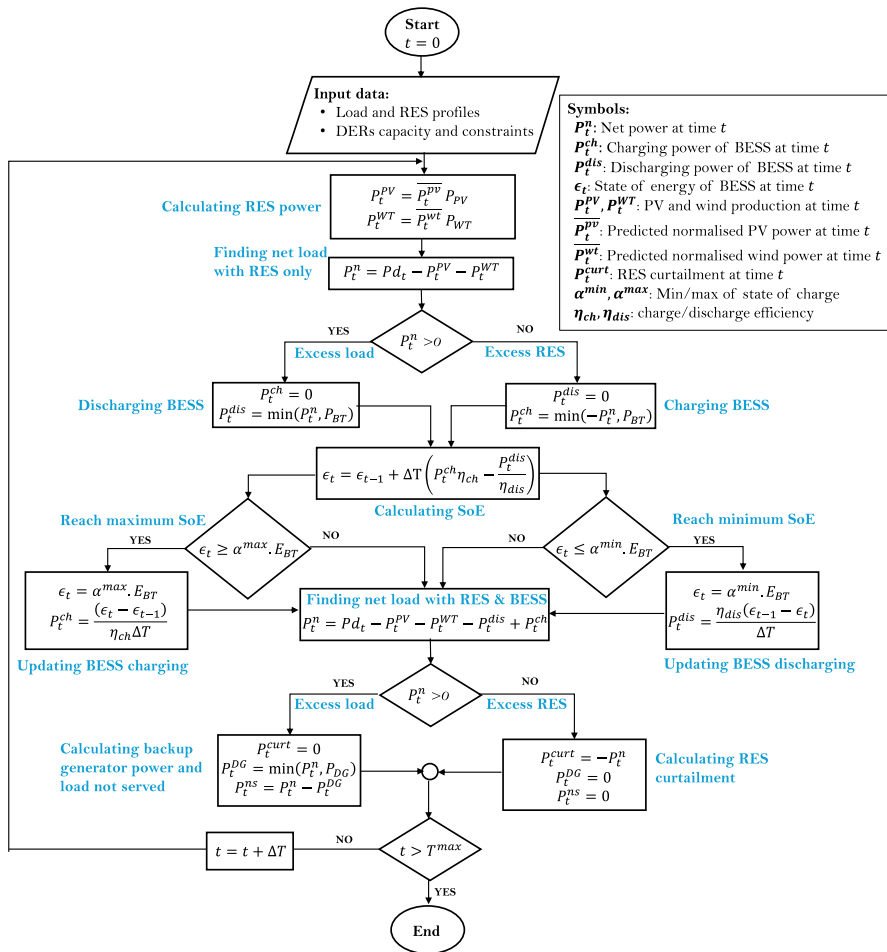


Fig. 2 The flowchart of the proposed rule-based MG scheduling model

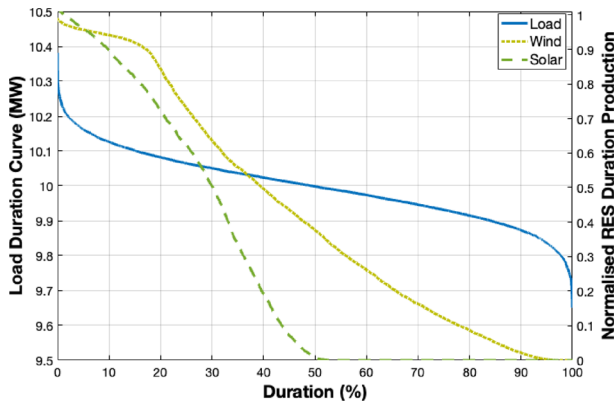


Fig. 3 Electricity load duration curves for the mine

4 Case study

This section presents the results of the proposed MG planning model for a case study based on a real-world, off-grid mine in Western Australia. The mine is located in a remote area with significant potential for solar and wind energy, showing capacity factors of around 43% for wind and 28% for solar, as shown by the normalised production-duration curves in Fig. 3. The electrical load demand follows a consistent 24/7 pattern, ranging from 9.5 to 10.5 MW, with an average of about 10 MW and a total annual energy of 87.5 GWh. The electricity demand profile is constructed from real data obtained from our case study, after excluding anomalies due to shutdowns or maintenance activities, based on consultations with mining engineers present at the site. This stable load profile reflects the high inflexibility that demands a continuous power supply. This may lead to substantially higher costs when developing a (near) zero-emission MG, which will be demonstrated in this section through comprehensive simulation studies.

For simulation studies, various input data are required, including cost and emission parameters, along with technical specifications for the energy sources. Detailed cost and emission values are available in [18], and a summary is

Table 1 Summary of the values used for the main cost and emission parameters

	Solar PV	Wind System	Lithium-ion (4 h, 0.25C)	Diesel Generator
Capital Cost (\$/kW)	1800	2200	2644	1200
Fixed O&M (\$/kW-yr)	45	60	7.57	52.5
Fuel Cost (\$/kWh)	–	–	–	0.17
Life Time (Years)	25	25	12	25
Operational Emissions (kg CO ₂ -e/GJ)	–	–	–	87.5

Table 2 Key Decision Variables for Pareto Solutions

Decision Variable	Symbol	Range
Installed Capacity of Wind Systems	P_{WT}	[0, 33 MW]
Installed Capacity of Solar PV Systems	P_{AC}^{PV}	[0, 33 MW]
Installed Capacity of BESSs (Power)	P_{BT}	[0, 33 MW]
Installed Capacity of BESSs (Energy)	E_{BT}	$[0.5P_{BT}, 4P_{BT}]$
Installed Capacity of Diesel Generators	P_{DG}	[0, 11 MW]

presented in Table 1. The key decision variables for the Pareto solutions, including the installed capacities of wind, PV, BESS (both power and energy), and diesel generators, are also summarised in Table 2. Diesel generators were chosen as the fossil-fuel source due to the lack of a natural gas pipeline at the mine site. The mine MG is assumed to have a 25-year lifespan, with a 3.1% compound interest rate. The maximum capacity of wind, PV, and BESS units is set at three times the peak demand (33 MW). To ensure maximum reliability, the ENS must be zero for all Pareto optimal solutions.

Simulation studies were carried out using Pymoo, a Python-based framework for multi-objective optimisation [19]. The aim of this case study was to carry out a comprehensive and rigorous comparative evaluation. The Pymoo library, which offers implementations of a wide range of state-of-the-art MOEAs, provided a suitable platform to facilitate this investigation.

The chosen algorithms - NSGA-II, NSGA-III, SMS-EMOA, U-NSGA-III, AGE-MOEA, and AGE-MOEA-II - represent a diverse range of established and contemporary MOEA strategies. MOEAs use different strategies to balance selection pressure, fitness assignment, and diversity preservation to approximate the Pareto-optimal front.

NSGA-II is a classic Pareto-based MOEA known for its simplicity and efficiency. It uses fast non-dominated sorting and crowding distance to maintain a diverse Pareto front without any user-defined parameters for diversity. NSGA-III introduces reference-point-based diversity preservation to maintain a well-distributed set of solutions by guiding the search towards a set of reference directions, thus overcoming the loss of selection pressure in high dimensions that NSGA-II experiences. U-NSGA-III embodies a unified strategy; the main innovation of this strategy is to avoid algorithm selection (or tuning) by the user. The algorithm itself ensures that it is using an appropriate mechanism (from a simple GA to NSGA-II to NSGA-III style) [20].

In contrast, SMS-EMOA represents indicator-based approaches, directly optimising the hypervolume contribution for selection, often leading to well-distributed solutions concentrated near the knee region of the Pareto front. This gives a strong convergence and diversity drive for low-dimensional fronts, at the expense of heavy computation as the number of objectives grows.

Finally, AGE-MOEA and its successor AGE-MOEA-II incorporate adaptive geometry estimation techniques, aiming to tailor the search process based on the evolving shape of the approximate Pareto front.

This thorough comparative methodology facilitates the identification of the most effective algorithmic strategies tailored to the distinct characteristics of the problem being examined. Hence, it reduces the likelihood of drawing biased conclusions favouring a particular algorithmic paradigm.

For consistency in all experiments, the population size was set at 100, and 100,000 function evaluations were performed for each algorithm. Each algorithm was run 31 times with different starting points, using varying seed values.

4.1 Numerical results

Before undertaking a comparison of the algorithms, the simulation outcomes for the NSGA-II algorithm are presented and analysed to enhance understanding of the solution space and Pareto fronts. The NSGA-II is used due to its wide application in the literature to solve similar problems in the realm of power system planning (e.g., [21–23]).

Figure 4 illustrates the Pareto solutions for three combinations of DER: 1) only diesel generator (DG: the star point), 2) without BESS (DG+RES: the square filled points), and 3) with BESS (DG+RES+BESS: the circle unfilled points). As shown in this figure, the addition of RES can significantly reduce both NPC and emissions. However, these reductions are saturated after a certain level of RES penetration. As expected, the addition of BESSs can significantly reduce emissions, but substantially increase the total cost. Hence, wind and solar PV systems without BESS generally present the most cost-effective solutions to reduce emissions by displacing fossil-fuel generation.

Expanding the capacity of the RES notably enhances energy curtailment, reaching up to 130 GWh annually, which is 50% more than the overall energy

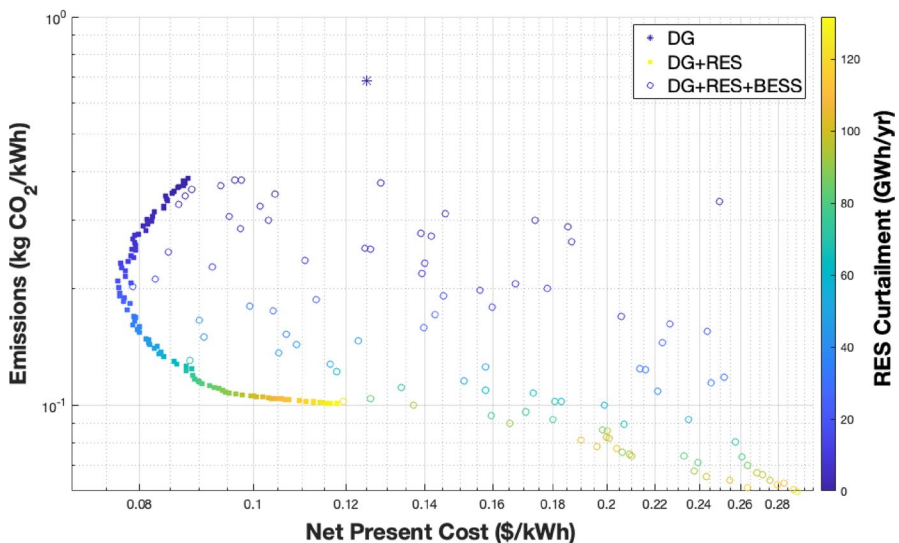


Fig. 4 The Pareto solutions for different DER combination cases

Table 3 Results of the selected solutions for different combinations of DERs

	Case (#)	RES (MW)	P _{BT} (MW)	E _{BT} (MWh)	P _{DG} (MW)	NPC (\$/kWh)	EM (kg CO ₂ /kWh)	REC (GWh/yr)
DG		0	0	0	10.5	0.125	0.685	0
DG+	(I)	23.7	0	0	9	0.077	0.209	19.1
RES	(II)	66	0	0	8.7	0.118	0.101	131.8
	(III)	10.3	0	0	9.4	0.088	0.384	0.0
DG+	(I)	19.9	1.2	1.2	9.5	0.084	0.248	8.7
RES+	(II)	66	24.5	33	8.5	0.290	0.060	125.7
BESS	(III)	10.8	1.1	2.2	9.3	0.097	0.380	0.0

demand. However, NPC does not change proportionally for scenarios without BESSs. Additionally, while BESSs have the potential to diminish energy curtailment to a degree, they cannot completely eliminate it, even at increased capacities, due to constraints on the allowable size of the BESS. For example, in cases with emissions below 0.1 kg of CO₂-e/kWh, energy curtailment exceeds 50 GWh/yr, even with the utilisation of BESS.

To perform a more detailed analysis, we have selected the best scenario from each of the three combinations of DER technologies, as shown in Table 3. In this table, Case numbers I, II, III represent the best solutions in terms of minimum NPC, GHG, and RES curtailment, respectively. For additional context, we have also included the base case, which relies solely on a diesel generator. This comparison offers deeper insights into the trade-offs between different objectives across the selected cases, enhancing the understanding of system performance under each scenario.

As shown in Table 3, for the base case (DG), NPC and emissions are 0.125 \$/kWh and 0.685 kg CO₂-e/kWh, respectively. However, the addition of RES (case I) can reduce the cost and emissions to 0.077 \$/kWh and 0.209 kg CO₂-e/kWh, respectively, representing 38% and 69% reductions in cost and emissions, respectively, compared to the base case. This case includes 19.1 GWh/yr of RES curtailment, which is 21% of the total demand. To minimise curtailment, the installed capacities of wind and PV systems should be decreased to the levels specified in Case III, resulting in an increase in both NPC and GHG emissions. Moreover, in DG+RES, the minimum GHG emissions (case II) are obtained when the installed capacity of wind and PV systems is three times the peak load demand. However, even with large amounts of RES curtailment in this case, the emissions cannot be reached below 0.101 kg CO₂-e/kWh due to periods of wind and/or sun drought.

Furthermore, the simulation results show that combining BESS with wind and/or solar PV systems, while having a diesel generator to fill in the gaps, can reduce emissions to as low as 0.060 kg CO₂-e/kWh, more than 90% and 40% reductions compared to the base case and case II of DG+RES, respectively. However, the NPC of this case is 0.290 \$/kWh, which is 32% and 45% more than the NPC of the base case and case II of DG+RES. However, using BESS reduces RES curtailment, even when it is as low as 10% of the load or 5% of the RES power rating.

4.2 Comparative analysis

This section compares the performance and the results obtained after executing the six evolutionary algorithms for both “DG+RES” and “DG+RES+BESS” combinations. The comparison between the Pareto sets and, consequently, between evolutionary algorithms is not straightforward and requires multi-dimensional analysis.

We first compare the algorithms in terms of the distance of their Pareto fronts with regard to the true Pareto set. Since the true Pareto sets are not known for our MG configurations, they are determined by comparing the final Pareto solutions of all algorithms and choosing the non-dominated solutions from this pool. The distance between Pareto fronts can be mathematically measured with different metrics, including the generational distance (GD) and the inverted generational distance (IGD) as well as their modified versions GD+ and IGD+ [24, 25].

The GD measures the average distance from solutions in the approximated Pareto front to the true Pareto front, while IGD does the reverse, calculating the mean distance from each point in the true Pareto front to the closest point in the approximated front. However, both fail to distinguish between dominated and non-dominated solutions. GD+ and IGD+ improve this process by penalising dominated solutions and focusing on non-dominated ones.

Table 4 presents the average values of the distance metrics for both the DG+RES and DG+RES+BESS combinations. The smallest values are achieved by the SMS-EMOA algorithm, followed by AGE-MOEA, which mostly offers the second smallest values for IGD and IGD+. SMS-EMOA stands out with the best overall performance across all metrics, although it has a significantly longer runtime compared to the other algorithms, particularly in the DG+RES combination, where it takes over seven times longer than the other algorithms. AGE-MOEA provides competitive results, with moderate performance and a shorter runtime.

Table 4 Multi-objective performance indicators

Case	ALG	GD	IGD	GD+	IGD+	HV	RunTime (min)
DG+RES	NSGA-II	0.0146	0.4561	0.0011	0.0037	1.9134	2.63
	NSGA-III	0.0121	1.4427	0.0018	0.0045	1.9047	2.8
	U-NSGA-III	0.0117	1.4675	0.0018	0.0046	1.9044	2.7
	SMS-EMOA	0.0010	0.3244	0.0001	0.0014	1.9294	19.11
	AGE-MOEA	0.0108	0.4403	0.0015	0.0027	1.9192	2.85
	AGE-MOEA-II	0.0072	1.6745	0.0014	0.0047	1.9043	3.01
DG+RES+BESS	NSGA-II	0.1361	0.4402	0.0863	0.0268	9.2435	159.68
	NSGA-III	0.0917	0.8167	0.0515	0.0220	9.4285	162.71
	U-NSGA-III	0.0880	0.8108	0.0485	0.0223	9.4146	162.47
	SMS-EMOA	0.0508	0.4035	0.0096	0.0151	9.5264	194.44
	AGE-MOEA	0.1096	0.5192	0.0700	0.0216	9.4245	160.95
	AGE-MOEA-II	0.1027	1.6780	0.0719	0.0232	9.2433	162.43

The bold and underlined values indicate the best performance among all compared methods for each evaluation metric

However, considering the capital cost of these infrastructures and the nature of the planning problems that do not need immediate results, SMS-EMOA is overall the best approach in all aspects. Note that in the DG+RES+BESS combination, the difference in run time is even less significant, which suggests that SMS-EMOA is indeed a well-rounded method for this application.

Additionally, while NSGA-II performs satisfactorily for the DG+RES combination, it shows higher values for the same metrics in the DG+RES+BESS combination, likely due to the increased number of decision variables and constraints. Moreover, AGE-MOEA-II, NSGA-III, and U-NSGA-III exhibit the highest values for these metrics, making them unsuitable options to solve this problem. This result was anticipated since they are primarily designed for many-objective problems with more than four objectives.

Moreover, Table 4 presents the hypervolume (HV) performance indicator. HV is a key metric that evaluates both the convergence and diversity of the Pareto front by calculating the volume of the region formed between the Pareto front solutions and a reference point. The reference point should be located outside the region and is typically defined in such a way that all Pareto-optimal solutions dominate it. Therefore, the method proposed in [26] is used here to select the suitable reference point to have a fair comparison. According to this method, the reference point can be calculated as follows:

$$\text{ref point} = (1 + 1/H) \cdot \max(obj_1, obj_2, \dots, obj_n) \quad (37)$$

where H is an integer value that is a function of the number of objectives (m) and the population size ($nPop$) as below.

$$C_{m-1}^{H+m-1} \leq nPop \leq C_{m-1}^{H+m} \quad (38)$$

Here, C_k^n represents the combination of k out of n . Using this formula and considering the population size of $nPop = 100$ for all algorithms, we obtain $11.651 \leq H \leq 12.4651$, so we select $H = 12$.

We observe from the table that the average HV is similar across all algorithms, with SMS-EMOA showing a slightly higher value, followed by AGE-MOEA and

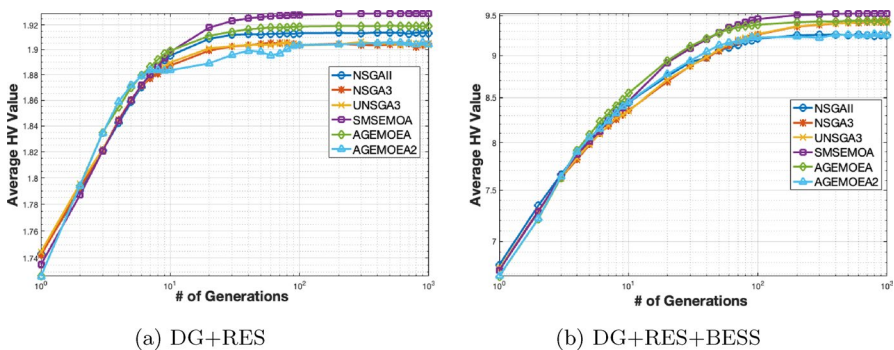


Fig. 5 Average values of HV over generations for DG+RES and DG+RES+BESS

NSGA-II. This observation is consistent with the previous comparison using distance-based metrics, such as GD, IGD, GD+, and IGD+.

Another important aspect of comparing the algorithms is their performance over generations. Since the MG planning problem is time-consuming, obtaining suitable Pareto solutions in fewer iterations is highly desirable. Figure 5 shows the average HV value of all six algorithms over the number of generations. As seen in the figure, for generations fewer than 10, all algorithms yield similar HV values, with the AGE-EMOA family algorithms slightly outperforming the others. However, for generations beyond 100, SMS-EMOA and AGE-MOEA begin to outperform the others, providing the highest HV values.

Moreover, when comparing the HV values obtained from DG+RES and DG+RES+BESS, we see a significant increase, 46%, in HV between generations 1 and 1000 for the DG+RES+BESS simulation study. In contrast, the change in the DG+RES combination is less than 10%. This difference is likely due to the higher number of decision variables in the case of DG+RES+BESS (5 variables) compared to the DG+RES (3 variables).

In terms of distance-based metrics, Figs. 6 and 7 provide further insight into the performance of the algorithms for both DER combinations. For the DG+RES combination, we observe that SMS-EMOA consistently achieves the lowest GD, GD+, IGD, and IGD+ values, indicating better convergence and distribution of the Pareto front compared to the other algorithms. AGE-MOEA and NSGA-II also demonstrate

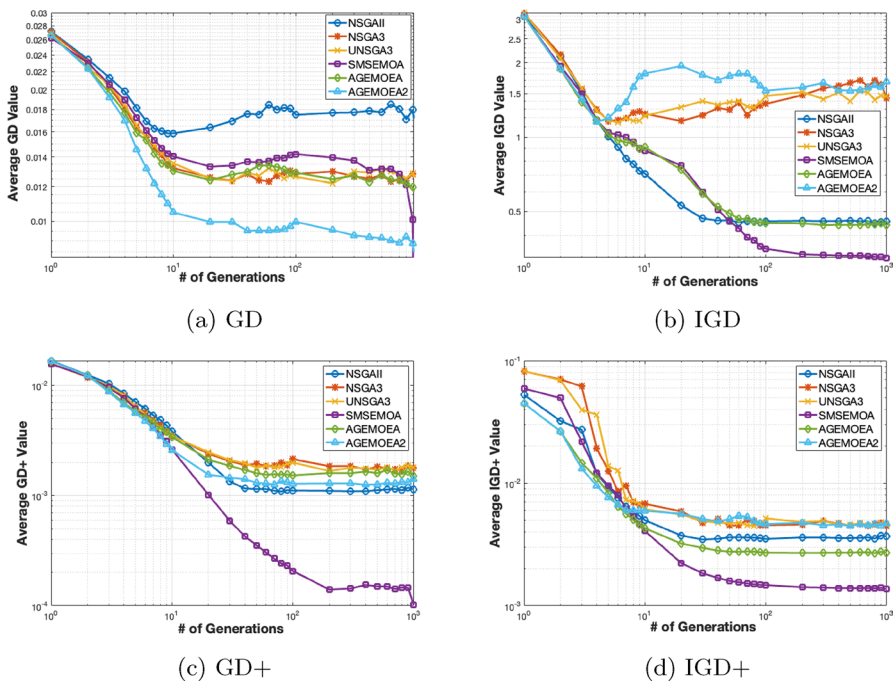


Fig. 6 Average values of distance metrics over generations for DG+RES combination

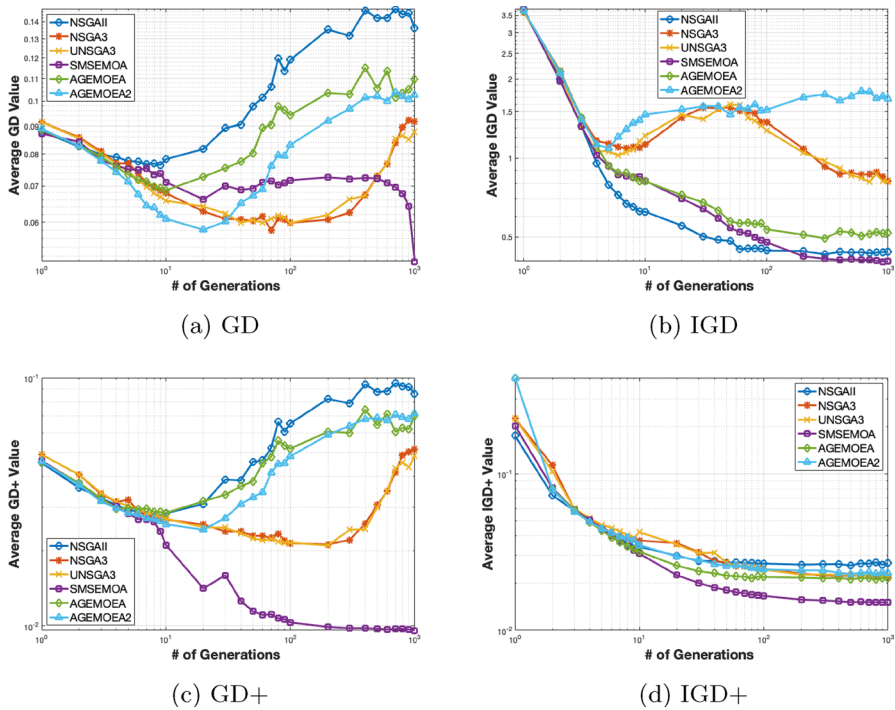


Fig. 7 Average values of distance metrics over generations for DG+RES+BESS combination

moderate performance, maintaining reasonable values, but with slightly higher distances compared to SMS-EMOA.

In the DG+RES+BESS combination, however, the gap between the algorithms becomes more pronounced. While SMS-EMOA still provides the best GD, IGD, GD+, and IGD+ values, the other algorithms, particularly NSGA-II, struggle with the increase in the number of decision variables. NSGA-II's performance noticeably degrades in the DG+RES+BESS combination, particularly for GD and GD+ metrics. This highlights the challenge posed by the increased complexity of the problem when BESS is included, making SMS-EMOA and AGE-MOEA more suitable for such cases, as they maintain lower distances, and thus better convergence and diversity in the Pareto front.

In general, although SMS-EMOA and AGE-MOEA perform robustly in both DER combinations, the DG+RES+BESS combination distinctly highlights the superior ability of these algorithms to handle more complex problems that involve a greater number of decision variables, as demonstrated by the improved distance metrics.

5 Conclusion and future works

In this paper, we proposed a multi-objective, two-stage optimisation framework for the optimal planning of renewable-based mining MGs. The framework aimed to minimise the total NPC, GHG emissions, RES curtailment, and improve system reliability. The study compared the performance of six advanced multi-objective evolutionary algorithms (NSGA-II, NSGA-III, U-NSGA-III, SMS-EMOA, AGE-MOEA, and AGE-MOEA-II) in finding Pareto-optimal solutions for two MG configurations: DG+RES and DG+RES+BESS. The results highlighted the advantages of using high-penetration RES in reducing emissions, with SMS-EMOA and AGE-MOEA showing the best overall performance in terms of convergence, diversity, and runtime efficiency. Moreover, the inclusion of BESS further reduced emissions, but led to higher NPC due to the increased cost of storage components. Moreover, the comparative analysis showed that SMS-EMOA consistently achieved the best results on multiple performance metrics, although it had a longer runtime, particularly for complex MG configurations.

Future research could incorporate frequency-constrained MG planning to improve stability and reliability, especially in systems with high RES integration. Exploring advanced evolutionary algorithms such as hybrid or decomposition-based methods can improve optimisation efficiency and accuracy. Moreover, linearising the battery degradation model would allow the use of mathematical programming methods, such as the traditional or augmented ϵ -constraint approaches for solving the multi-objective problem. Additionally, applying AI techniques, such as reinforcement learning and neural networks, can optimise MG operations in real-time, reducing computational time and improving adaptability to changing conditions. Combining these approaches with stochastic and robust optimisation will address uncertainties and further enhance the resilience of renewable-based microgrids.

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Data and code availability The datasets and code used in this study are available on GitHub and can be accessed via the following repository: <https://bit.ly/48xfm6X>. The repository includes all the necessary data and source code for reproducing the results presented in the paper. We encourage the research community to use and modify the materials under the terms of the open-source repository licence.

Declarations

Conflict of interest The authors of this paper declare that they have no Conflict of interest.

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